



Robotic Exoskeletons: From Fiction To Reality

Exoskeletons are ideal for use in almost every industry, including medical, agriculture, automobile, construction, manufacturing, and so on. These can help people maintain fitness and prevent health issues as well as help those who face hardships in performing daily tasks



Exoskeletons moving from fiction to reality (Credit: www.likeablestem.com)

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From old science-fiction movies like *Aliens* to the new ones like *Avengers*, it has been common in films to feature fascinating robotic exoskeletons. But their availability in real life is much more usual than one would notice.

The term exoskeleton literally means an external skeleton structure wherein the skeletal system is outside an organism's body, as seen in arthropods like spiders, insects, and crustaceans. Borrowing and translating this idea, just like many other innovations, the military was the first to combine robotics and biomechanics to create wearable frameworks that could be

worn on a specific body part, or even on the entire body, to boost a human being's performance.

Unlike an autonomous robot, it does not replace the worker altogether. Neither does it work like a prosthetic that replaces the original amputated body part. According to a report from MarketsandMarkets, the exoskeleton market is predicted to grow to 2,810.5 million dollars by 2023.

Types of exoskeletons

There can be many types of exoskeletons based on different classifications. Depending on the powering mechanism,